COMPLETE WORK PLAN

Game-Theoretic Analysis of Misinformation, Trust, and Moderation Failure

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0. FOUNDATIONAL CLARITY (START HERE)

Objective

Establish what this paper is, and just as importantly, what it is not.

Tasks

- Finalize the **core claim**:
  Emotionally distorted payoffs + evolving trust cause structural failure of moderation, even with learning and enforcement.
- Clearly define what the paper **does not attempt**:
  - No real-world prediction
  - No empirical fitting
  - No psychological realism
  - No equilibrium solving
- Identify target research audience:
  - Computational social science
  - Game theory (dynamic / behavioral)
  - Trust & safety / platform governance

Output

- Frozen scope statement
- One-sentence contribution statement

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1. THEORY & CONCEPTUAL FRAMEWORK (MOST IMPORTANT)

Purpose

Build a **mechanistic theory** explaining why moderation fails, independent of simulations.

1.1 Core Concepts (define clearly)

- **Perceived payoff vs objective payoff**
- **Emotion as payoff distortion** (not internal feelings)
- **Trust as an endogenous dynamic state**
- **Intervention as a strategic social signal**
- **Adaptive adversary as a strategic player**

1.2 Causal Mechanism

- Emotional amplification inflates short-term reward of belief-aligned sharing
- Punishment and correction introduce delayed or identity-threatening costs
- Trust rescales how these costs are perceived
- Trust erosion flips incentive signs
- Adversary exploits emotional amplification and trust decay

1.3 Informal Propositions (no proofs)

- When emotional distortion exceeds corrective cost, misinformation sharing is rational
- Punishment reduces misinformation only above a trust threshold
- Adaptive adversaries prevent stable convergence

1.4 Theoretical Gap

- Existing models assume:
    - Stable incentives
    - Fixed legitimacy
    - Trust as background or homophily
- This paper removes those assumptions

Output

- 2–3 pages of Conceptual Framework / Theory section
- Explicit qualitative predictions

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2. LITERATURE POSITIONING

Purpose

Position the paper **above** existing work, not beside it.

Tasks

- Read core papers from:
  - Opinion dynamics & backfire
  - Fake news games
  - RL-based moderation
  - Emotion & virality models
- Build a **comparison matrix**:
  - What agents optimize
  - How emotion is modeled
  - Whether trust is endogenous
  - How interventions are treated
  - Failure modes captured
- Write synthesis (not summaries):
  - What they explain
  - What they cannot explain
  - Where this paper fits

Output

- Literature comparison table
- Related Work section draft

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3. MODEL FORMALIZATION

Purpose

Translate theory into a **minimal formal game**.

3.1 Players

- Regular users
- Platform / moderator
- Adaptive misinformation adversary

3.2 State Variables

- Belief (continuous)
- Trust (continuous)
- Emotional susceptibility

3.3 Actions

- Share / not share
- Intervene / not intervene
- Adversarial content injection choices

3.4 Payoff Structure

- Belief alignment reward
- Emotion-amplified social reward
- Trust-scaled correction or punishment cost

3.5 Trust Dynamics

- Trust decreases under perceived hostility
- Trust recovery is slower than decay
- Intervention accelerates erosion under conflict

3.6 Adversary Adaptation

- Observes trust and polarization
- Adjusts emotional intensity and targeting
- Objective: maximize polarization, minimize trust

Output

- Model section (text + light math)
- Explicit statement: no equilibrium analysis

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4. SIMULATION DESIGN

Purpose

Study **dynamic trajectories** of the game.

Tasks

- Define environment:
  - 50–200 agents
  - Fixed initial network
  - Discrete time
  - Small stochastic noise
- Define intervention regimes:
  1. No intervention
  2. Punishment-heavy moderation
  3. Trust-aware / non-punitive intervention
- Define metrics:
  - Average trust over time
  - Belief variance (polarization)
  - Misinformation prevalence
  - Time to collapse / recovery

Output

- Experimental Setup section

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5. IMPLEMENTATION

Purpose

Execute the minimal model cleanly.

Tasks

- Implement:
  - Agent class
  - Environment loop
  - Trust updates
  - Intervention module
  - Adversary module
- Run simulations:
  - Baseline
  - Each intervention regime
  - Multiple seeds
  - Parameter sweeps
- Generate plots:

    - Trust vs time
    - Polarization vs time
    - Regime comparisons

Output

- Reproducible simulation results
- Clean figures

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6. RESULTS & FAILURE MODE ANALYSIS

Purpose

Explain **what fails, how, and why**.

Tasks

- Describe trajectories (early vs late behavior)
- Identify failure modes:
    - Short-term suppression / long-term collapse
    - Trust erosion spiral
    - Adversarial escalation loop
    - Non-convergence or cycling
- Compare intervention regimes:
    - What “works better” actually means
    - Why nothing fully solves the problem

Output

- Results section
- Failure mode taxonomy

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7. DISCUSSION

Purpose

Tie everything back to theory and implications.

Tasks

- Explain why punishment fails structurally
- Explain why learning and optimization fail
- Discuss implications for platform design:
    - Preserving trust > enforcing truth
    - Slow interventions outperform aggressive ones

Output

- Discussion section

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8. LIMITATIONS & ETHICS

Purpose

Preempt reviewer attacks.

Tasks

- Explicitly state limitations:
    - No real data
    - No prediction
    - Minimal agents
    - No psychological modeling
- Justify simulation approach
- Ethical framing:
    - No behavioral prescriptions
    - No manipulation guidance

Output

- Limitations section

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9. WRITING, POLISH, SUBMISSION

Tasks

- Assemble full paper

- Ensure logical flow and consistent terminology
- Polish figures and captions
- Final reference and formatting checks
- Submission-ready manuscript

FINAL NOTE (IMPORTANT)

This is a **theory-first, game-theoretic paper with simulation support**.
If Sections 1–3 are strong, everything else becomes execution.